The Input Side Algorithm

Figure 6-5 shows a simple convolution problem: a 9 point input signal, $x[n]$, is passed through a system with a 4 point impulse response, $h[n]$, resulting in a $9 + 4 - 1 = 12$ point output signal, $y[n]$. In mathematical terms, $x[n]$ is convolved with $h[n]$ to produce $y[n]$. This first viewpoint of convolution is based on the fundamental concept of DSP: decompose the input, pass the components through the system, and synthesize the output. In this example, each of the nine samples in the input signal will contribute a scaled and shifted version of the impulse response to the output signal. These nine signals are shown in Fig. 6-6. Adding these nine signals produces the output signal, $y[n]$.

Let's look at several of these nine signals in detail. We will start with sample number four in the input signal, i.e., $x[4]$. This sample is at index number four, and has a value of 1.4. When the signal is decomposed, this turns into an impulse represented as: $1.4\delta[n-4]$. After passing through the system, the resulting output component will be: $1.4h[n-4]$. This signal is shown in the center box of the nine signals in Fig. 6-6. Notice that this is the impulse response, $h[n]$, multiplied by 1.4, and shifted four samples to the right. Zeros have been added at samples 0-3 and at samples 8-11 to serve as place holders. To make this more clear, Fig. 6-6 uses *squares* to represent the data points that come from the shifted and scaled impulse response, and *diamonds* for the added zeros.

Now examine sample $x[8]$, the last point in the input signal. This sample is at index number eight, and has a value of -0.5. As shown in the lower-right graph of Fig. 6-6, $x[8]$ results in an impulse response that has been shifted to the right by eight points and multiplied by -0.5. Place holding zeros have been added at points 0-7. Lastly, examine the effect of points $x[0]$ and $x[7]$. Both these samples have a value of zero, and therefore produce output components consisting of all zeros.

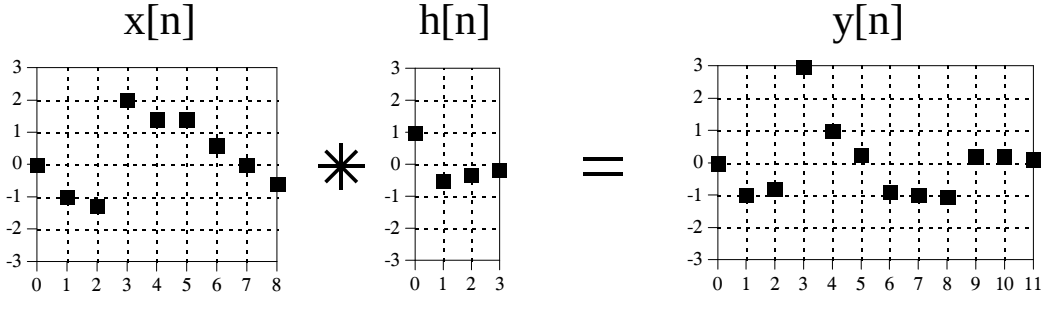

FIGURE 6-5
Example convolution problem. A nine point input signal, convolved with a four point impulse response, results in a twelve point output signal. Each point in the input signal contributes a scaled and shifted impulse response to the output signal. These nine scaled and shifted impulse responses are shown in Fig. 6-6.